

# Periodic Labour Force Survey (PLFS) Concepts

Course ID: MOSPI-C005 | Domain: Labour Statistics | Level: Beginner | Duration: 6 hours

## PROTOTYPE LEARNING MATERIAL NOTICE

This learning module is compiled from curriculum standards for role-aligned training demonstration.

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## 1. Introduction

The Periodic Labour Force Survey (PLFS) is India's flagship household survey for measuring labour market indicators. Launched by the National Statistical Office (NSO), MoSPI in April 2017, PLFS replaced the earlier quinquennial Employment-Unemployment Surveys (EUS) conducted by the erstwhile NSSO.

PLFS provides estimates of key labour force indicators on two time frequencies:

- **Annual estimates**: For both rural and urban areas (July-June reference period)
- **Quarterly estimates**: For urban areas only (January-March, April-June, July-September, October-December)

This dual-frequency design enables timely monitoring of urban labour markets while maintaining comprehensive annual coverage of both rural and urban areas.

## 2. Survey Design

### 2.1 Sample Design

PLFS uses a stratified multi-stage design:

- First Stage Units (FSUs): Census villages (rural) and Urban Frame Survey (UFS) blocks (urban)
- Second Stage Units: Households
- Stratification: Rural areas stratified by 4 MPCE classes; Urban areas stratified by town size class.

Sample size: ~1.02 lakh households per year (about 7,200 FSUs, with urban FSUs revisited quarterly for the panel design).

### 2.2 Rotating Panel in Urban Areas

Urban FSUs are divided into 4 sub-samples (rotation groups). Each sub-sample is surveyed in one quarter and then re-surveyed in the same quarter of the following year. This panel structure enables quarter-on-quarter and year-on-year comparison for urban labour market dynamics.

## 2.3 Reference Period for Employment

PLFS uses multiple reference periods:

- **Usual Activity (Principal + Subsidiary)\*\***: 365 days preceding the date of survey
- **Current Weekly Status (CWS)\*\***: 7 days preceding the date of survey
- **Current Daily Status (CDS)\*\***: Each day of the reference week (7 days)

The choice of reference period affects measured employment and unemployment rates significantly.

## 3. Key Concepts and Definitions

### 3.1 Labour Force

The labour force comprises all persons who are either employed or unemployed (seeking and available for work) during the reference period. Labour Force = Employed + Unemployed.

### 3.2 Worker Population Ratio (WPR)

$$\text{WPR} = (\text{Number of persons employed} / \text{Population}) \times 100$$

Also called Employment Rate. Measures the proportion of the population that is employed. A high WPR can reflect both genuine employment and necessity-driven employment (e.g., distress employment in agriculture).

### 3.3 Labour Force Participation Rate (LFPR)

$$\text{LFPR} = (\text{Labour Force} / \text{Population}) \times 100$$

Measures the proportion of the population that is either employed or actively seeking employment. Low LFPR among women and youth is a key structural feature of India's labour market.

### 3.4 Unemployment Rate (UR)

$$\text{UR} = (\text{Unemployed} / \text{Labour Force}) \times 100$$

Measures the proportion of the labour force that is without work but seeking and available for work. India's unemployment rate is relatively low by international standards but disguised unemployment (underemployment) is significant.

### 3.5 Usual Principal Activity (UPS)

The activity in which a person spent the major time during the 365 days preceding the survey date. Used for classifying persons into broad categories: workers, unemployed, and out-of-labour-force (students, domestic duties, etc.).

### 3.6 Usual Subsidiary Activity (USS)

Activity pursued for a minor part of the year in addition to the principal activity. Adding USS to UPS gives the "usual activity (principal + subsidiary)" or UPSS measure, which captures a broader segment of workers, particularly seasonal agricultural workers.

### 3.7 Current Weekly Status (CWS)

Employment status determined based on activities during the 7 days preceding the survey date. A person is employed (CWS) if they worked for at least one hour on at least one day during the reference week.

### 3.8 Current Daily Status (CDS)

Determined for each day of the reference week. A person can be employed for part of a day and unemployed for the rest. CDS measures person-days of work and unemployment. CDS unemployment rate is highest among the three measures as it captures underemployment.

### 3.9 Industry and Occupation Classification

Industries classified using National Industrial Classification (NIC) 2008 (aligned with ISIC Rev. 4). Occupations classified using National Classification of Occupations (NCO) 2015 (aligned with ISCO-08).

## 4. Key Indicators Produced by PLFS

| Indicator                | Definition                              | Reference Period |
|--------------------------|-----------------------------------------|------------------|
| LFPR                     | Labour Force / Population × 100         | UPS, CWS, CDS    |
| WPR                      | Employed / Population × 100             | UPS, CWS, CDS    |
| Unemployment Rate        | Unemployed / LF × 100                   | UPS, CWS, CDS    |
| Employment by industry   | Distribution across NIC sectors         | Usual activity   |
| Employment by occupation | Distribution across NCO categories      | Usual activity   |
| Employment by status     | Self-employed, regular/salaried, casual | Usual activity   |
| Wages                    | Wage/earnings by category, sex, sector  | Current status   |

## 5. Comparison: PLFS vs. Earlier EUS

| Feature              | Earlier NSSO EUS             | PLFS                                     |
|----------------------|------------------------------|------------------------------------------|
| Frequency            | Quinquennial (every 5 years) | Annual (rural+urban) + Quarterly (urban) |
| Design               | Cross-sectional              | Rotating panel (urban)                   |
| Sample size          | ~1 lakh HHs (large round)    | ~1 lakh HHs/year (continuous)            |
| Urban quarterly data | Not available                | Available from PLFS                      |
| Reference period     | UPS, CWS, CDS                | UPS, CWS, CDS (same)                     |
| Latest base          | 2011-12                      | 2017-18 (first full year)                |

## 6. Practical Example: Computing Unemployment Rate

From a hypothetical PLFS sample for urban areas (CWS reference period):

Population (15+ years): 1,00,000 persons  
 Employed (CWS): 42,000  
 Unemployed (seeking + available): 3,000  
 Out of Labour Force (students, domestic duties, etc.): 55,000

Labour Force = 42,000 + 3,000 = 45,000  
 $LFPR = (45,000 / 1,00,000) \times 100 = 45.0\%$   
 $WPR = (42,000 / 1,00,000) \times 100 = 42.0\%$   
 $Unemployment\ Rate = (3,000 / 45,000) \times 100 = 6.7\%$

## 7. Data Quality Considerations

- **Reporting bias**: Respondents may under-report casual or part-time work, especially women's work in domestic enterprises.
- **Reference period sensitivity**: UPS, CWS, and CDS measures give different results; comparisons across surveys must use the same reference period.
- **Seasonal variation**: Labour market conditions in agriculture vary significantly by season; annual averages may mask within-year fluctuations.
- **Urban-rural comparability**: The rotating panel design is only for urban areas; rural estimates are cross-sectional only.
- **Non-response**: Non-response in follow-up rounds of the panel can introduce attrition bias.
- **Informal employment**: Self-employed and casual workers are harder to classify; "contributing family workers" are counted as employed even with zero earnings.

## 8. Key Learning Points

1. PLFS replaced the quinquennial NSSO EUS from 2017-18; provides annual (rural+urban) and quarterly (urban) estimates.
2. The three reference periods (UPS/UPSS, CWS, CDS) give different employment and unemployment rates.
3. Labour force = employed + unemployed (seeking + available).
4. LFPR measures labour market participation; WPR measures employment; UR measures unemployment among the labour force.
5. Urban quarterly estimates use a rotating panel; rural estimates are cross-sectional.
6. CDS unemployment rate is highest as it captures part-day underemployment.

## 9. Learning Outcomes

On completing this material, you should be able to:

- Describe the PLFS survey design, including sample design and rotating panel for urban areas.
- Define and compute LFPR, WPR, and unemployment rate under different reference periods.
- Distinguish between UPS, UPSS, CWS, and CDS measures and explain when each is appropriate.
- Compare PLFS with the earlier NSSO Employment-Unemployment Survey.
- Identify data quality considerations in interpreting PLFS estimates.